GRAVITYSORT · FEASIBILITY ASSESSMENT

GravitySort feasibility assessment

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline), closed at TRL 3 against the STEM check, GVS-CAL-001 v0.9, the build plan and the BOM

GravitySort feasibility assessment

GravitySort is feasible on paper as a workshop-built, pedal-driven concentrator: every physical claim in the STEM check sits within known science or is plausible pending a test, the constructable design closes with welding, drilling and bought parts, and the main open questions are performance (fine-gold retention, table ratio) and two safety and mass margins that only testing or Amish's decisions can close. Method: STANDARDS section 23. Evidence: GVS-STM-001, GVS-CAL-001, GVS-BLD-001, `bom/bom.csv`, `cad/fidelity.json`.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design (physics)	Bowl speed, G, settling, fluidization head, power, burst, critical speed, braking and tipping all check by first-order calculation; five details failed and were fixed under the standing instruction of 2026-10-23 (drum height, filter, buffer rating, gold balance, the climb to the rings found by the independent review) and one by the detailed CAD (head gussets)	High	Fine-gold retention in the bed (STM S3; upflow near minimum fluidization of 75 um grains) and table ratio (S15) need tests; critical speed on the real frame 1,274 to 1,976 rpm (S8); full-tank sway near the table frequency (S33, O3); pedal power at 80 G exceeds 60 W (PR-006)	GVS-STM-001; GVS-CAL-001 sections 2 to 11
Safety	Guards, lid interlock, disc brake with parking latch, speed display and a burst safety factor of 16 at sprint speed; stands on 13.1 degrees with a full tank	Medium	Containment of a liner fragment is shown face first but not edge first (PR-016 at risk; steel band option to Amish, GVS-DEC-001 O2); guard openings against ISO 13857 not yet checked (REG G5)	GVS-CAL-001 section 10; GVS-RSK-001
Materials	Mild steel tube and plate, HDPE drums, PU liner on GFRP, bought bearings and bicycle parts	Medium	PU liner life on quartz unknown (PR-014); hand-laid GFRP quality decides the burst margin; painted steel in a wet splash zone needs upkeep	GVS-STM-001 S7, S29
Manufacturing and assembly	Every part has a making sketch and an assembly step; no lathe; D3 model with every fastener and weld passes the fidelity check for all three options	High	Casting the liner on a six-segment printed core is untried (STM S25); used bevel gearbox hard to source (quarter-turn belt option in DEC)	GVS-BLD-001; `cad/fidelity.json`
Power and controls	Pedal baseline (48 W at 60 G) with the MotionCore 250 W option (5.2 times margin); no electronics of its own beyond a bicycle computer	High	Rider effort at 80 G; motor pack energy 0.55 kWh per shift is the user's choice	GVS-CAL-001 section 7
Water	1.19 m3/h, gravity-fed from a 1.7 m post, just meeting the proposed 10 kPa net at every ring at mid-tank with 3/4 in valve and rotameter (F6); filter protects the 1.0 mm holes; flow at the low end of commercial practice (O5)	Medium	Sites without an upstream source need a pump (site rule decided 2026-10-02); filter must be flushed when the rotameter flow drops	GVS-CAL-001 sections 4 and 6
Supply of parts	Steel, bicycle parts, V-belts, bearings, HDPE drums and irrigation fittings are sold in mining towns; the liner resin and rotary union are specialist	Medium	Union seal drag and price unconfirmed; resin supply for PU casting to check in Colombia	`bom/bom.csv`; GVS-DEC-001 items to confirm
Cost	$594 in parts against the $455 value-engineering target	Medium	USD 139 over the target; savings listed in GVS-DEC-001 (quarter-turn belt, used bearings, bushes)	GVS-CAL-001 section 12; GVS-MKT-001
Mass and transport	99.6 kg in six loads, heaviest 27.3 kg (estimate)	Low	0.4 kg under the 100 kg of PR-011; the model's solid volume	

Aspect	Assessment	Rating	Risks and notes	Evidence
			gives more (see GVS-CAL-001 section 11 note), so R11 is effectively at risk until parts are weighed	GVS-CAL-001 section 11; `cad/fidelity.json`

Ratings: High, Medium, Low.

Verdict

Feasible to build and test as a TRL 4 prototype once Amish decides the open items in GVS-DEC-001 (concept selection O1, containment band O2). No aspect is rated infeasible.